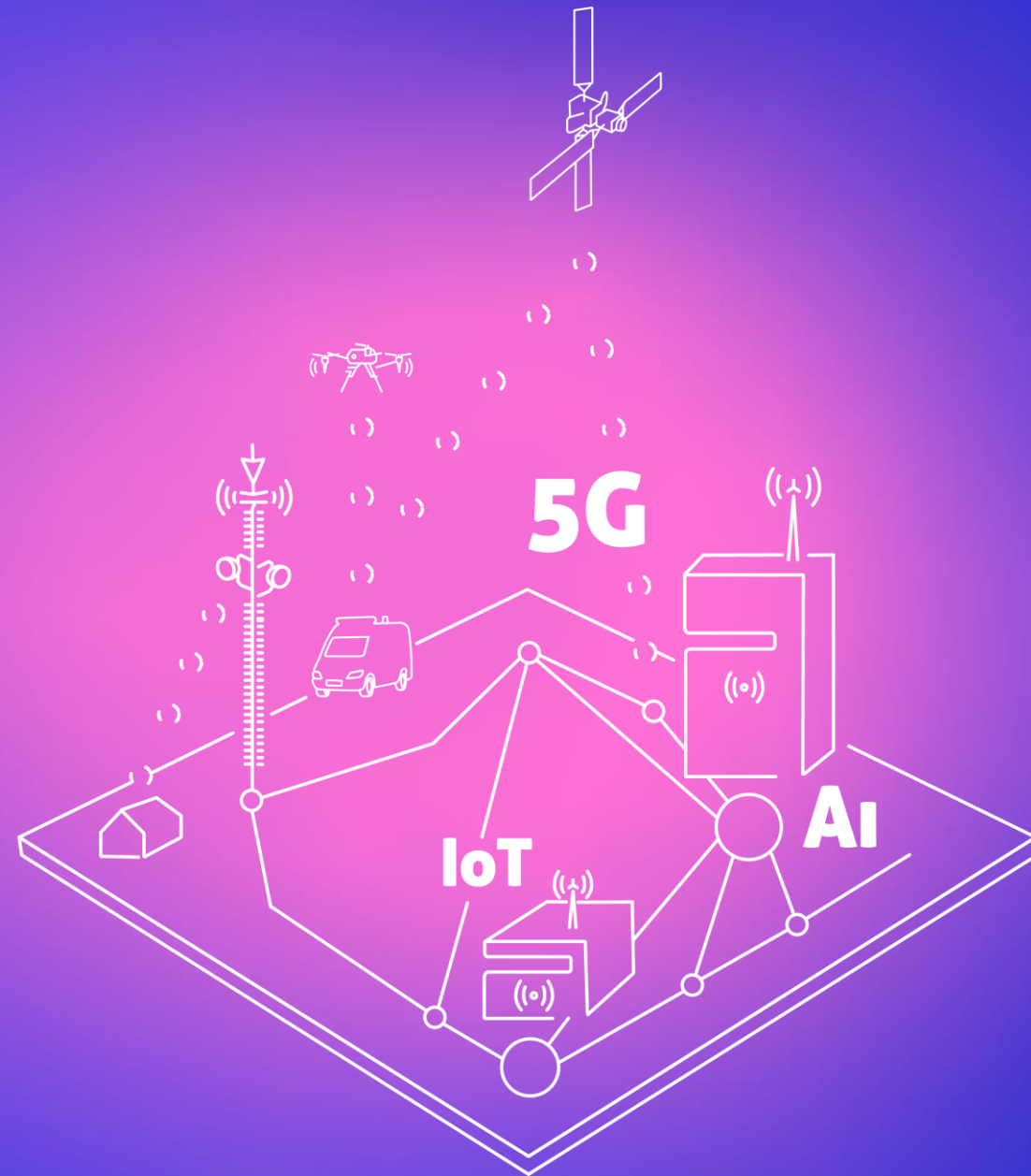




5. Prediction models

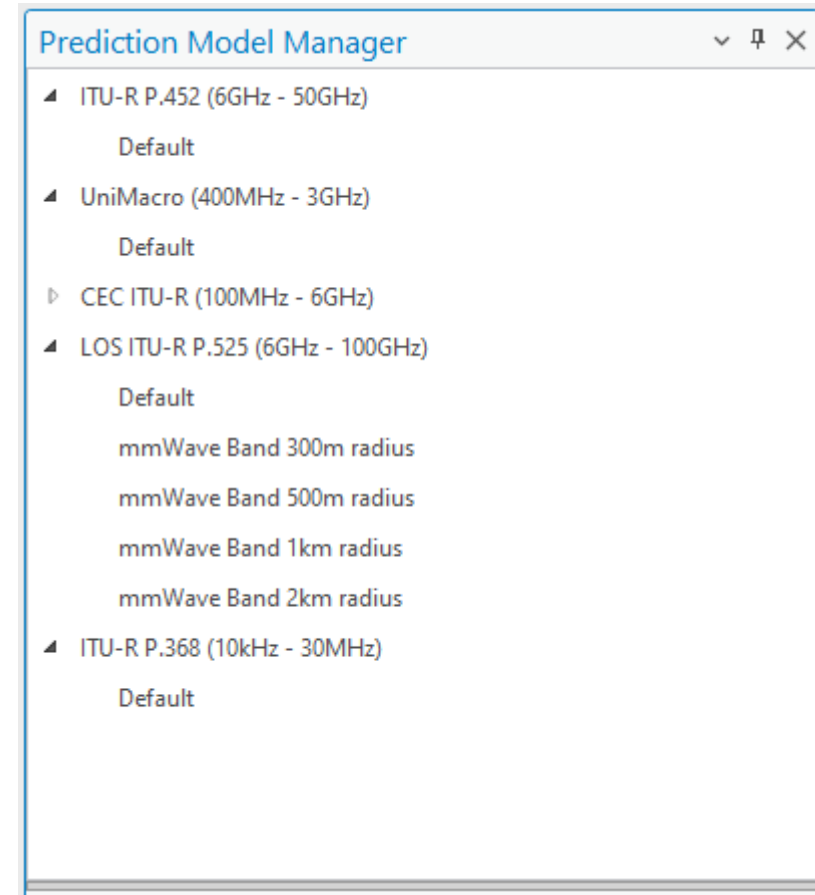
www.cellular-expert.com



$$\text{Field Strength} = \text{EIRP} - \text{Antenna Attenuation} - \text{Path Loss}$$

Prediction Models

- ITU-R P.452 (6GHz to 50GHz)
- UniMacro (400MHz to 3GHz)
- CEC ITU-R (100MHz to 6GHz)
- LOS ITU-R P.525 (6GHz to 100 GHz)
- ITU-R P.368 (10kHz to 30MHz)



CE Path Loss models (10kHz - 100 GHz)

1. CEC ITU-R Model (100MHz – 6GHz) is a combination model intended for use in a variety of different radiocommunication systems which is derived explicitly from ITU-R path loss modelling methods as follows:
 - a. Receive antenna in LOS condition – path loss calculated as FSL based on Recommendation ITU-R P.525 ([ref URL](#));
 - b. Receive antenna in OLOS condition – total path loss modelled as a combination of basic FSL calculated based on Recommendation ITU-R P.525 ([ref URL](#)) and clutter loss calculated based on Recommendation ITU-R P.2108 ([ref URL](#));
 - c. Receive antenna in NLOS condition – path loss as a combination of basic FSL calculated based on Recommendation ITU-R P.525 ([ref URL](#)), additional losses due to diffraction calculated based on Recommendation ITU-R P.526 ([ref URL](#)) and the clutter losses calculated based on Rec. ITU-R P.2108 ([ref URL](#)).
2. ITU-R P.452 Model (6GHz – 50GHz) is provided as a universally applicable model with very wide frequency range from 0.1-50 GHz. Its implementation is based on the methodology described in the Recommendation ITU-R P.452 ([ref URL](#)). This model does not provide for definition of OLOS visibility condition; instead it considers clutter as part of general obstacles category and accordingly distinguishes only two radio visibility cases:
 - a. Receive antenna in LOS condition – path loss modelled based on FSL principle;
 - b. Receive antenna in NLOS condition – total path loss modelled using a combination of basic transmission losses and losses due to diffraction.
3. LOS ITU-R P.525 Model (6GHz – 100GHz) is the FSL path loss calculated based on method in Recommendation ITU-R P.525 ([ref URL](#)). As such it could be used for modelling of radio links where LOS is considered a necessary condition, e.g., for Fixed (Point-to-Point) Links or Mobile Systems in mmWave bands.
4. UniMacro Model (400MHz – 3GHz) is the CE's proprietary combination model developed over the years of practical experience with the operational planning of cellular mobile networks in the frequency ranges from 400-2600 MHz. It had been fine tuned to produce coverage predictions that are most closely aligned with what could be expected to be experienced by the actual mobile network users in the field. The model will model different path losses depending on radio visibility conditions as follows:
 - a. Receive antenna in LOS condition – path loss modelled based on FSL principle;
 - b. Receive antenna in OLOS condition – path loss modelled using Extended Hata (Open Area) model with additional clutter loss calculated based on Recommendation ITU-R P.2108 ([ref URL](#));
 - c. Receive antenna in NLOS condition – path loss modelled using Extended Hata model with additional losses due to diffraction calculated based on Recommendation ITU-R P.526 ([ref URL](#)) as well as clutter losses based on Rec. ITU-R P.2108 ([ref URL](#)).
5. ITU-R P.368 (10kHz – 30MHz)

Prediction Models. Default

Prediction Model Manager

ITU-R P.452 (6GHz - 50GHz)

Default

UniMacro (400MHz - 3GHz)

CEC ITU-R (100MHz - 6GHz)

LOS ITU-R P.525 (6GHz - 100GHz)

ITU-R P.368 (10kHz - 30MHz)

Prediction Model Manager

ITU-R P.452 (6GHz - 50GHz)

UniMacro (400MHz - 3GHz)

CEC ITU-R (100MHz - 6GHz)

Default

1km radius

2km radius

3km radius

5km radius

7km radius

10km radius

15km radius

20km radius

25km radius

30km radius

50km radius

100km radius

Small Cell 200m radius

Small Cell 300m radius

WiFi 200m radius

WiFi 300m radius

WiFi 500m radius

LOS ITU-R P.525 (6GHz - 100GHz)

Prediction Model Manager

ITU-R P.452 (6GHz - 50GHz)

UniMacro (400MHz - 3GHz)

CEC ITU-R (100MHz - 6GHz)

LOS ITU-R P.525 (6GHz - 100GHz)

Default

mmWave Band 300m radius

mmWave Band 500m radius

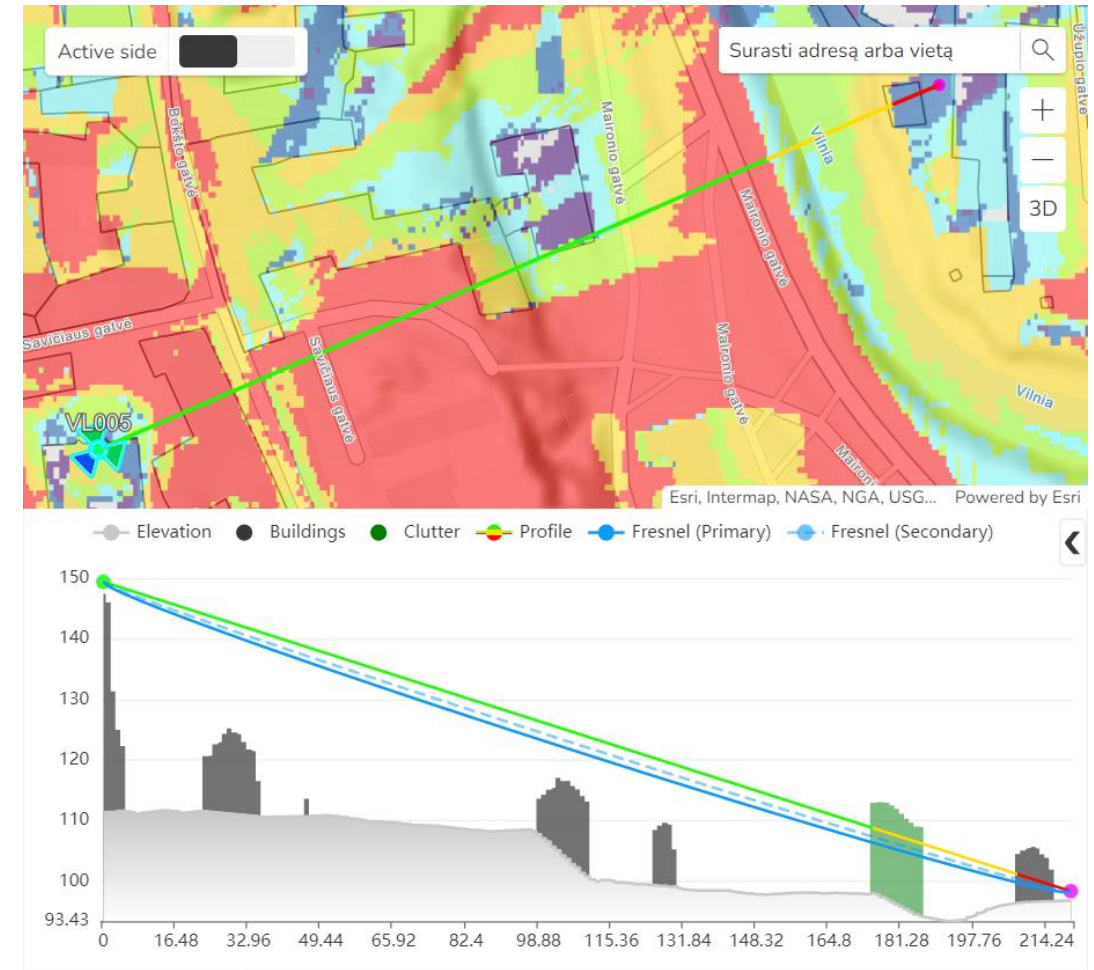
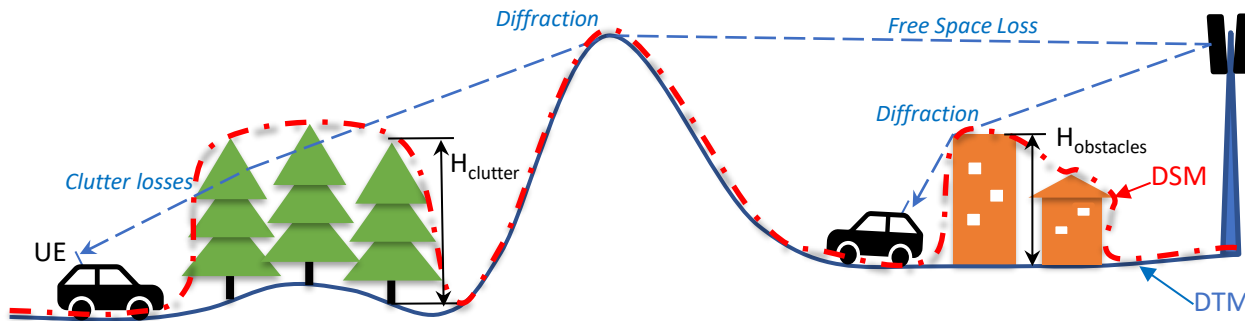
mmWave Band 1km radius

mmWave Band 2km radius

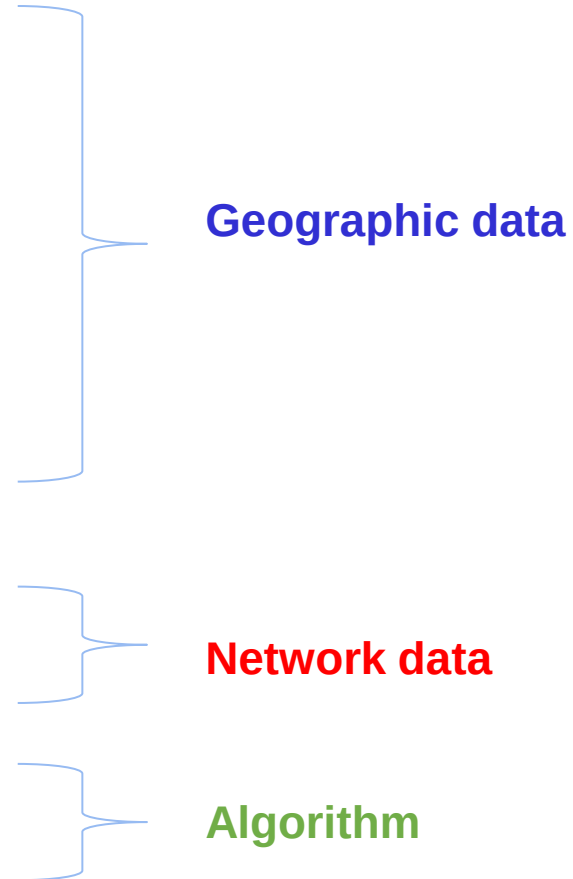
ITU-R P.368 (10kHz - 30MHz)

CEC ITU-R (30MHz – 6GHz)

- For frequencies from about 30 MHz to about 6 GHz.
- Modelling:
 - LOS
 - OLOS
 - NLOS



- Elevation
- Clutter classes*
- Clutter height grid*
- Receiver settings
- Prediction model settings



* *Optional*

Path loss equation

Path loss in dB:

$$L = k_{\text{off}} + k_{\log D} \cdot \log(d) + k_{\log F} \cdot \log(f)$$

Offset coefficient (K_{Off}) - Constant offset (dBm). Default value 32

Distance coefficient ($K_{\log D}$) - Distance influence coefficient. Default value 20

Frequency coefficient ($K_{\log F}$) - Frequency influence coefficient. Default value 20

Offset coefficient, dB:

32

Distance coefficient:

20

Distance coefficient obstructed:

30

Frequency coefficient:

20

Path loss equation

Path loss in dB:

$$L = k_{off} + k_{\log D} \cdot \log(d) + k_{\log F} \cdot \log(f)$$

Offset coefficient (K_{Off}) - Constant offset (dBm). Default value 32

Distance coefficient obstructed (K_{LogD}) - Distance influence coefficient. Default value 30

Frequency coefficient (K_{LogF}) - Frequency influence coefficient. Default value 20

Offset coefficient, dB:

32

Distance coefficient:

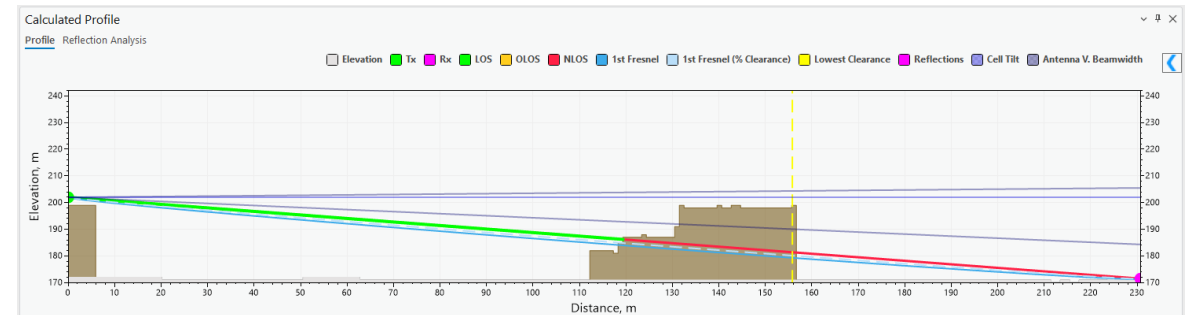
20

Distance coefficient obstructed:

30

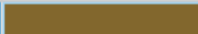
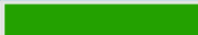
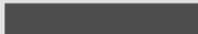
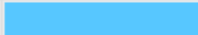
Frequency coefficient:

20



- Diffraction loss for solid obstacle:
 - Building clutter class
 - Elevation
- Clutter loss
 - Based on diffraction calculation
 - P.2108 Clutter Loss
- Penetration loss (Outdoor – Indoor)
- Receiver loss

▼ Clutter class: Buildings

Clutter name	Color
Buildings	
Forest	
Dense forest	
Urban	
Dense urban	
Bare ground	
Crops	
Water	
Road	

Nominal distance, m:

Diffraction loss coefficient:

Penetration loss offset, dB:

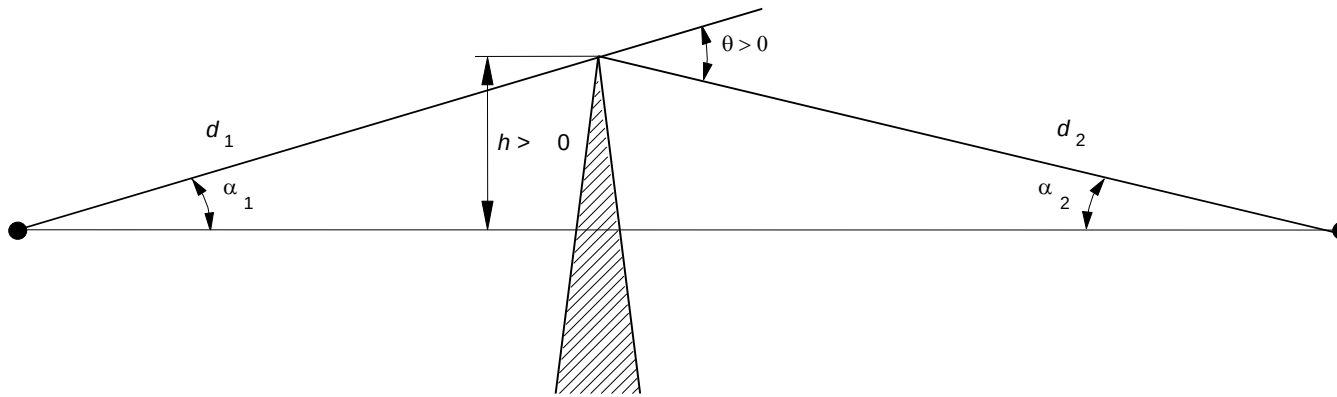
Penetration loss distance coefficient:

Penetration loss frequency coefficient:

Receiver point loss offset, dB:

Rec. ITU-R P.526

Idealized model of diffraction over a single obstruction.



P.2108 Clutter Loss Estimation

- Method 1: Additional clutter shadowing loss with diffraction as dominant effect (section 3.1)

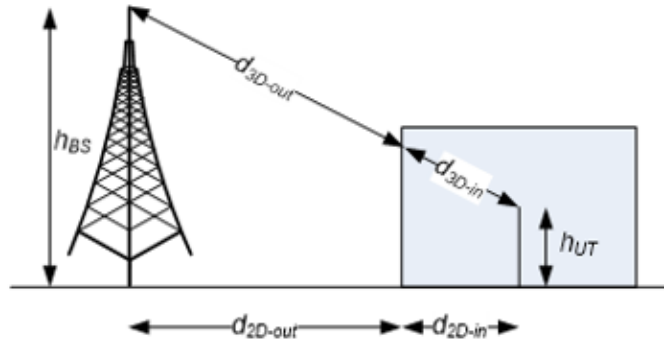
Input parameters:				
Frequency, GHz (0,03-3 GHz):	2.1			
Antenna height (agl), m:	1.5			
Urban scenario street width, m:	27 (default value 27 m)			
Representative clutter height, m:	10			
Clutter clearance angle, deg:	17.47477426			
v :	6.04019561			
$J(v)$:	28.45751028			
Adjusted $J(v)$:	28.45751028			
Clutter loss for open area scenarios:		22.4 dB		
Clutter loss for obstructed area scenarios:		19.6 dB		

clutter

Field:	Selection:	Rows:			
OBJECTID *	clutter_class_name	clutter_class_id	height	nominal_distance	
1	1	water	1	0	27
2	2	trees	2	20	27
3	3	flooded vegetation	4	3	27
4	4	crops	5	2	27
5	5	built area	7	2	27
6	6	bare ground	8	0	27
7	7	snow	9	0	27
8	8	rangeland	11	3	27
Click to add new row.					

Penetration loss (Outdoor – Indoor)

CE Outdoor to Indoor Path Loss calculation is realised based on method recommended in 3GPP TR 38.901 ([ref URL](#)). This method accounts for indoor portion of the total radio signal propagation path as shown in picture:



Definition of indoor propagation path in 3GPP TR 38.901

For general purpose modelling of typical building entry losses, two types of loss profiles are assumed:

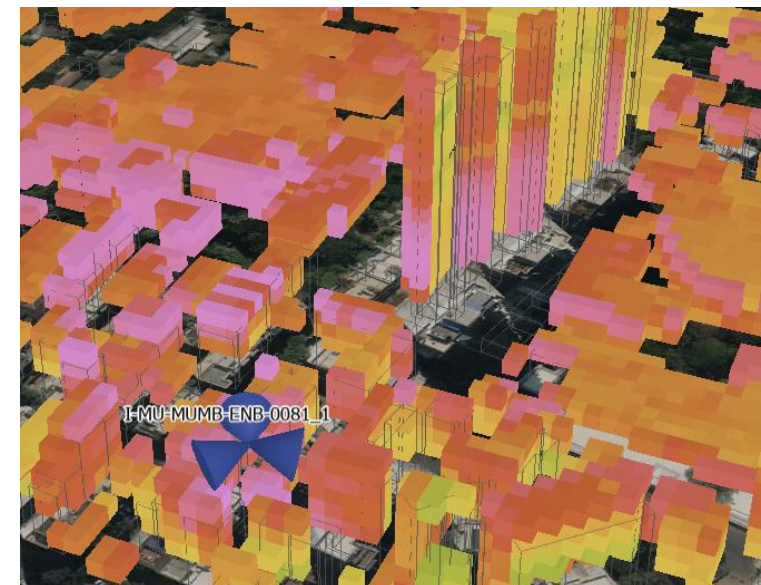
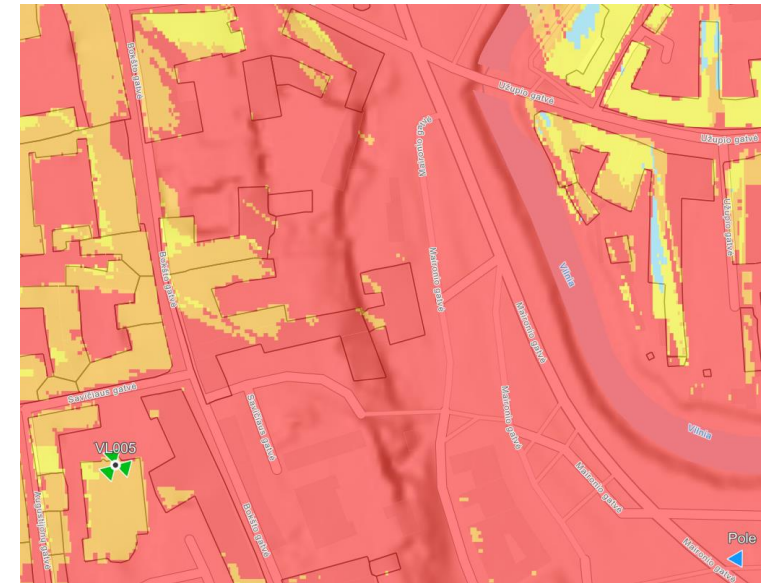
- Low-loss BEL Model assumes a wall penetration losses characteristic of average traditional buildings;
- High-loss BEL Model assumes a wall penetration losses characteristic of modern thermally insulated buildings.

The corresponding BEL and building penetration losses are calculated as follows:

	Path loss through external wall: PL_{tw} in [dB]	Indoor loss: PL_{in} in [dB]	Standard deviation: σ_P in [dB]
Low-loss model	$5 - 10 \log_{10} \left(0.3 \cdot 10^{\frac{-L_{glass}}{10}} + 0.7 \cdot 10^{\frac{-L_{concrete}}{10}} \right)$	$0.5 d_{2D-in}$	4.4
High-loss model	$5 - 10 \log_{10} \left(0.7 \cdot 10^{\frac{-L_{IRglass}}{10}} + 0.3 \cdot 10^{\frac{-L_{concrete}}{10}} \right)$	$0.5 d_{2D-in}$	6.5

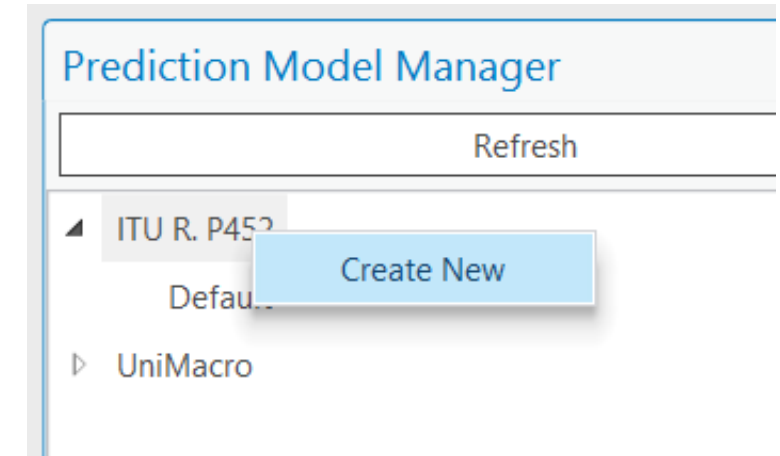
Where:

$$L_{glass} = 2 + 0.2f \quad L_{concrete} = 5 + 4f \quad L_{IRglass} = 23 + 0.3f \quad f - \text{frequency in GHz.}$$



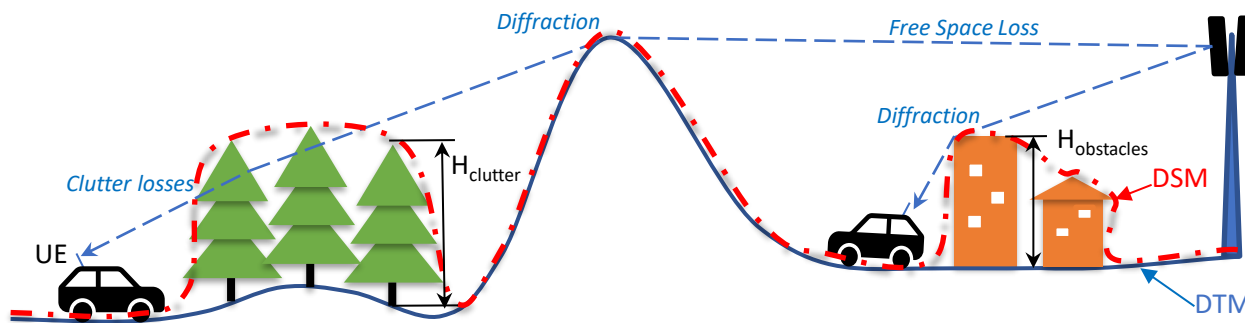
Prediction model manager

- Cellular Expert tab > Prediction Model Manager
- Default can not be deleted and it takes parameters from it for new models

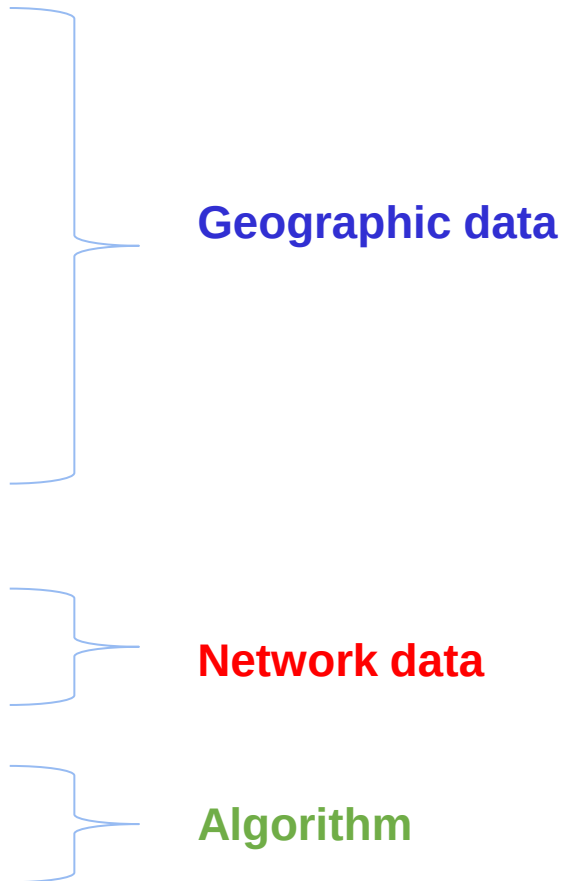


LOS ITU-R P.542 (6GHz – 50GHz)

- For frequencies from about 6 GHz to about 100 GHz



Radius, km:	5
Effective Earth radius, km:	8500
Time Percentage (%):	0.01
Receiver height, m:	1.5
Offset coefficient, dB:	32
Distance coefficient:	20
Frequency coefficient:	20
Use Multipath Focusing:	No

- Elevation
 - Clutter classes*
 - Clutter height grid*
 - Receiver settings
 - Prediction model settings
- 
- A diagram on the right side of the slide uses blue curly braces to group the input data items into three categories. The first brace groups 'Elevation', 'Clutter classes*', and 'Clutter height grid*', with the label 'Geographic data' in blue text to its right. The second brace groups 'Receiver settings', with the label 'Network data' in red text to its right. The third brace groups 'Prediction model settings', with the label 'Algorithm' in green text to its right.
- Geographic data**
- Network data**
- Algorithm**

* *Optional*

Path loss equation

Path loss in dB:

$$L = k_{\text{off}} + k_{\log D} \cdot \log(d) + k_{\log F} \cdot \log(f)$$

Offset coefficient (K_{Off}) - Constant offset (dBm). Default value 32

Distance coefficient ($K_{\log D}$) - Distance influence coefficient. Default value 20

Frequency coefficient ($K_{\log F}$) - Frequency influence coefficient. Default value 20

Offset coefficient, dB:

32

Distance coefficient:

20

Distance coefficient obstructed:

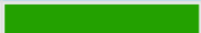
30

Frequency coefficient:

20

- Diffraction loss
 - Building clutter class
 - Elevation
- Penetration loss (Outdoor – Indoor)

▼ Clutter class: Buildings

Clutter name	Color
Buildings	
Forest	
Dense forest	
Urban	
Dense urban	
Bare ground	
Crops	
Water	
Road	

Penetration loss offset, dB:

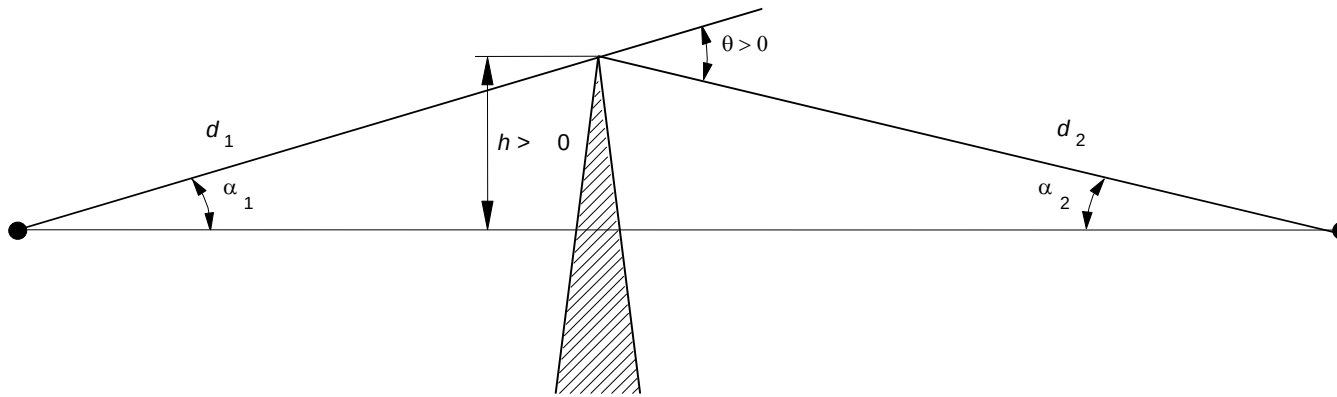
Penetration loss distance coefficient:

Penetration loss frequency coefficient:

Single Knife Edge Diffraction

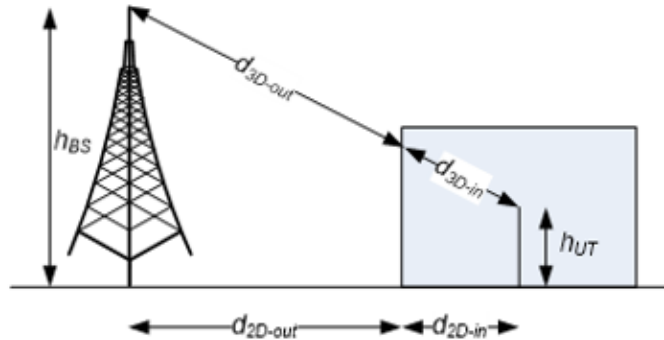
Rec. ITU-R P.526

Idealized model of diffraction over a single obstruction.



Penetration loss (Outdoor – Indoor)

CE Outdoor to Indoor Path Loss calculation is realised based on method recommended in 3GPP TR 38.901 ([ref URL](#)). This method accounts for indoor portion of the total radio signal propagation path as shown in picture:



Definition of indoor propagation path in 3GPP TR 38.901

For general purpose modelling of typical building entry losses, two types of loss profiles are assumed:

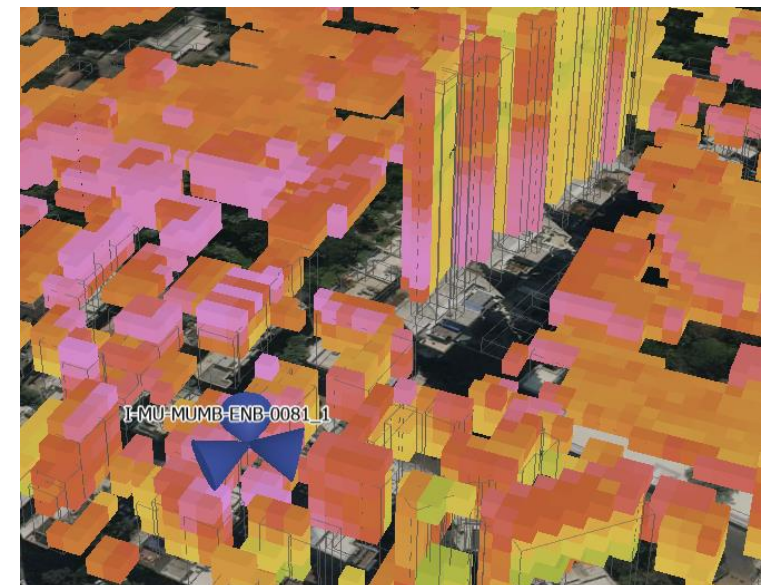
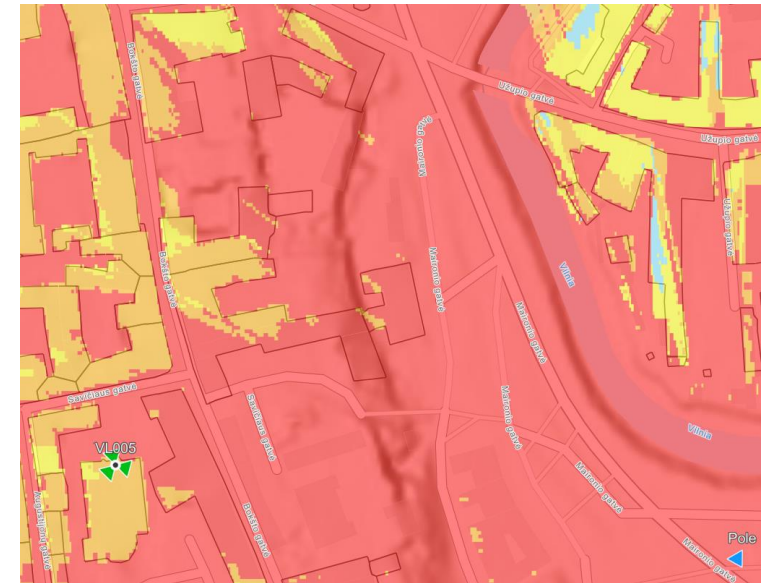
- Low-loss BEL Model assumes a wall penetration losses characteristic of average traditional buildings;
- High-loss BEL Model assumes a wall penetration losses characteristic of modern thermally insulated buildings.

The corresponding BEL and building penetration losses are calculated as follows:

	Path loss through external wall: PL_{tw} in [dB]	Indoor loss: PL_{in} in [dB]	Standard deviation: σ_P in [dB]
Low-loss model	$5 - 10 \log_{10} \left(0.3 \cdot 10^{\frac{-L_{glass}}{10}} + 0.7 \cdot 10^{\frac{-L_{concrete}}{10}} \right)$	$0.5 d_{2D-in}$	4.4
High-loss model	$5 - 10 \log_{10} \left(0.7 \cdot 10^{\frac{-L_{IRglass}}{10}} + 0.3 \cdot 10^{\frac{-L_{concrete}}{10}} \right)$	$0.5 d_{2D-in}$	6.5

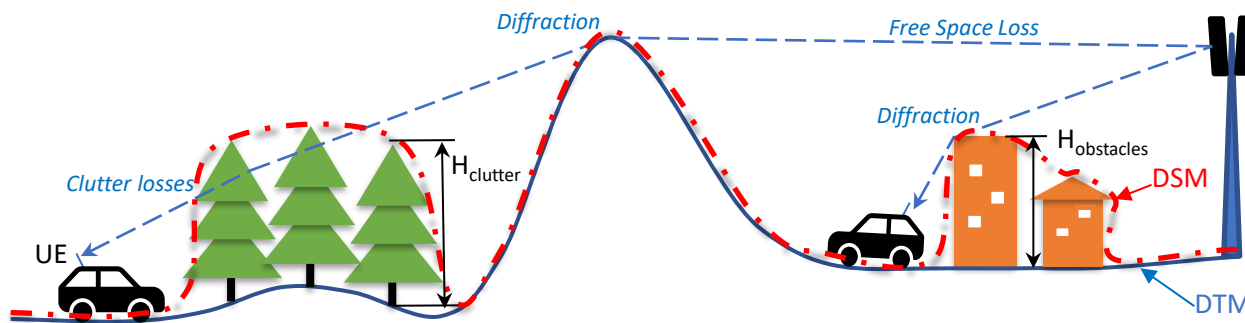
Where:

$$L_{glass} = 2 + 0.2f \quad L_{concrete} = 5 + 4f \quad L_{IRglass} = 23 + 0.3f \quad f - \text{frequency in GHz.}$$



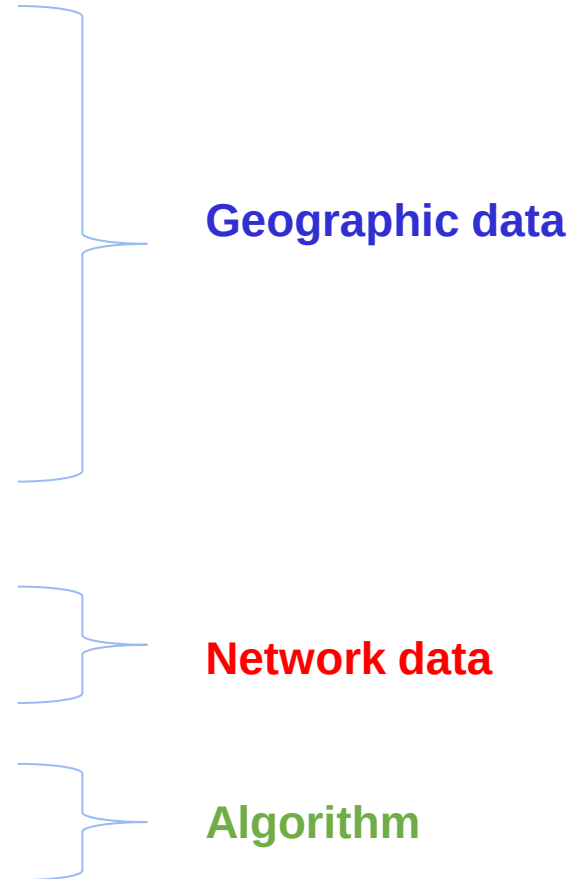
LOS ITU-R P.525 (6GHz – 100GHz)

- For frequencies from about 6 GHz to about 100 GHz



Configuration name:	Default
Radius, km:	5
Effective Earth radius, km:	8500
Receiver height, m:	1.5
Offset coefficient, dB:	32
Distance coefficient:	20
Frequency coefficient:	20

- Elevation
- Clutter classes*
- Clutter height grid*
- Receiver settings
- Prediction model settings



* *Optional*

Path loss equation

Path loss in dB:

$$L = k_{\text{off}} + k_{\log D} \cdot \log(d) + k_{\log F} \cdot \log(f)$$

Offset coefficient (K_{Off}) - Constant offset (dBm). Default value 32

Distance coefficient ($K_{\log D}$) - Distance influence coefficient. Default value 20

Frequency coefficient ($K_{\log F}$) - Frequency influence coefficient. Default value 20

Offset coefficient, dB:

32

Distance coefficient:

20

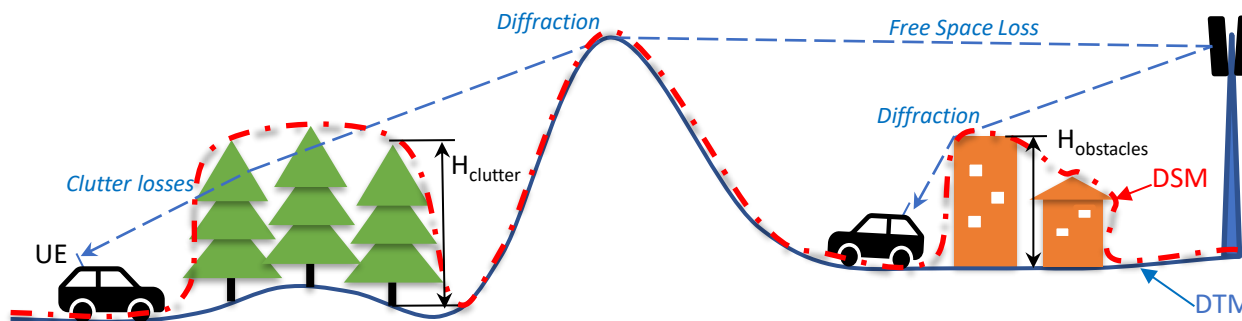
Distance coefficient obstructed:

30

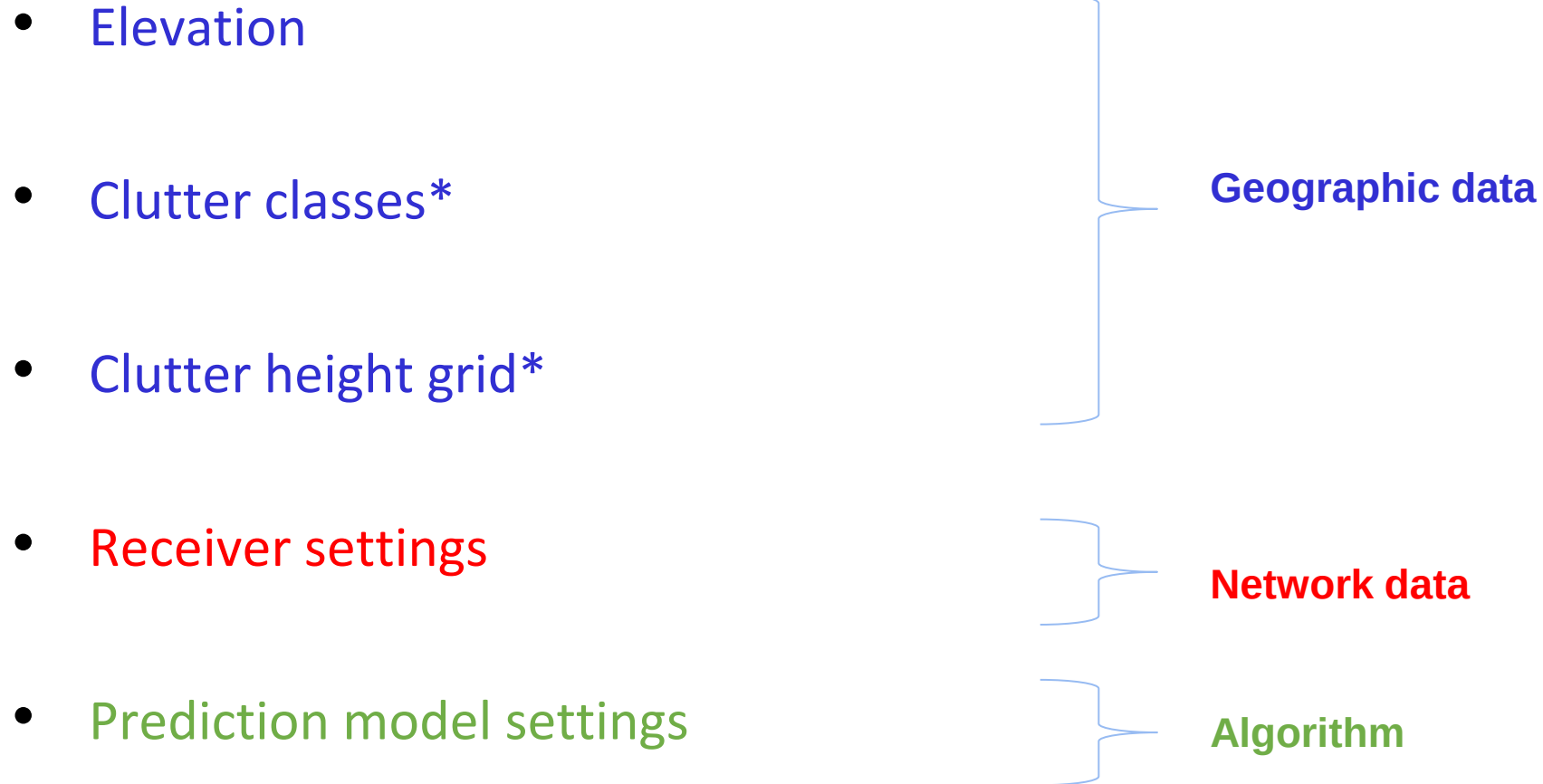
Frequency coefficient:

20

- Frequency: ~ 100 MHz - 2 GHz (3 GHz)
- Distance: up to 100 km
- 9999 Model (Ericsson)



Model Name:	Default
Maximum Radius (km):	2
Effective Earth Radius:	8500
Constant Receiving Antenna Height Above DTM (m):	1.5
Free Space Loss KOff:	32
Free Space Loss KLogD:	20
Free Space Loss KLogF:	20
Hata Loss: A0	36.2
Hata Loss: A1	30.2
Hata Loss: A2	-12
Hata Loss: A3	0.1
Use Line of Sight:	Yes
Use High Building Entry Loss:	No



* *Optional*

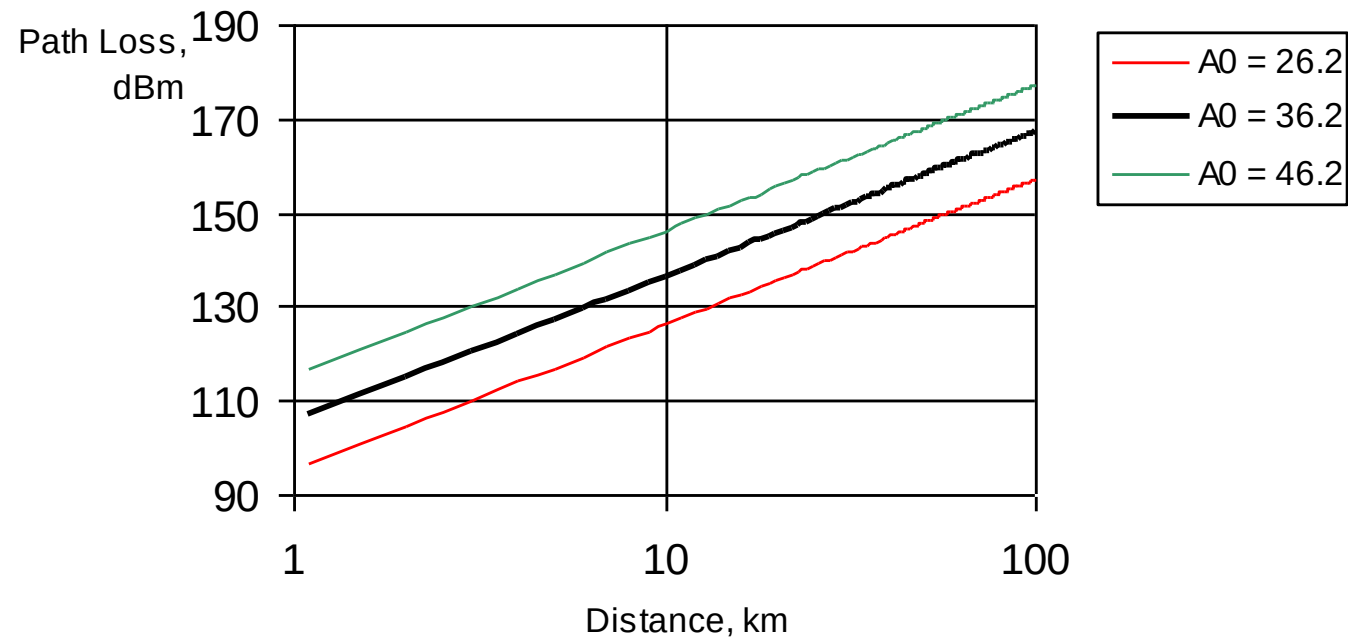
- Line-Of-Sight Model Loss
- 9999 Ericsson
- Single Knife Edge Diffraction

Path Loss Equation: 9999 Ericsson

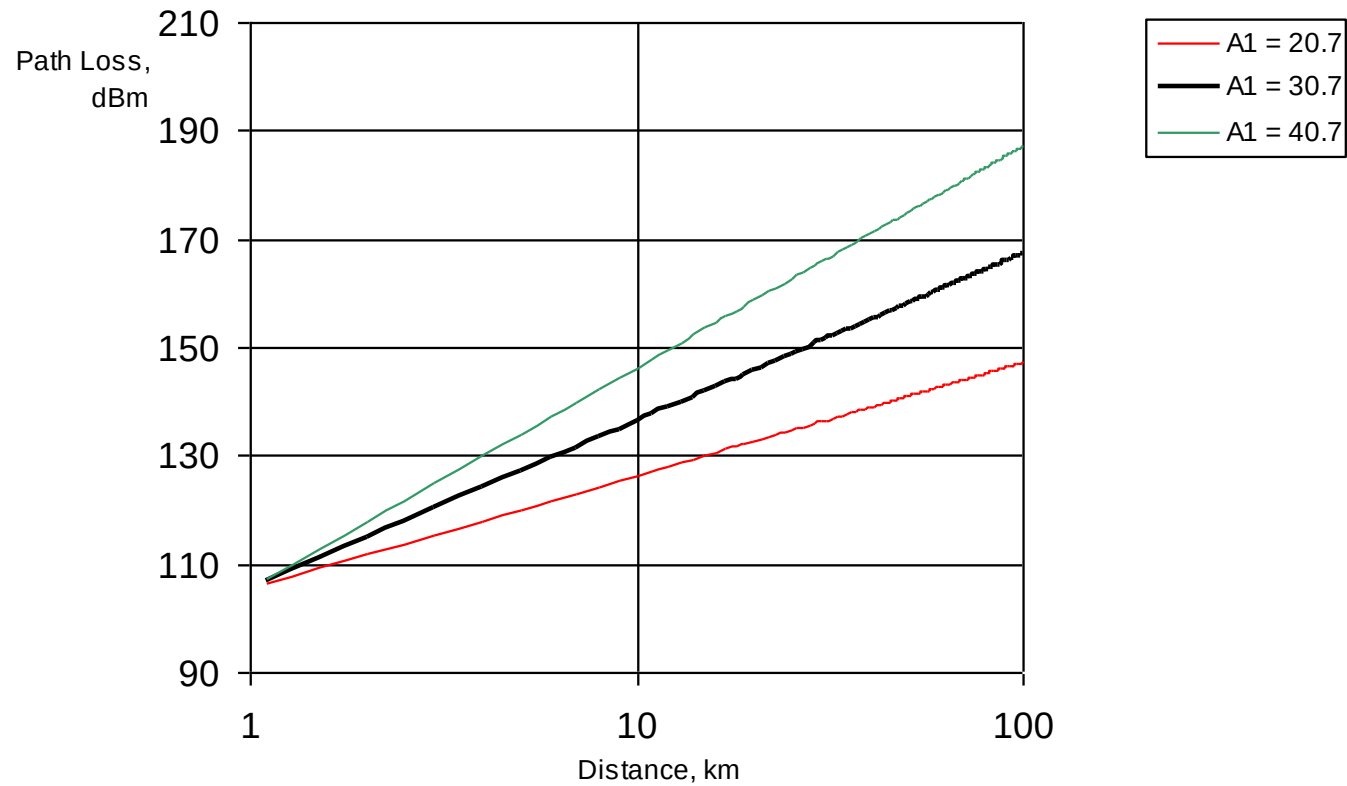
Path loss in dB:
$$L_H = a_0 + a_1 \cdot \log(d) + a_2 \cdot \log(h_B) + a_3 \cdot \log(h_B) \cdot \log(d) + 3.2 \cdot (\log(11.75 \cdot h_M))^2 + g(f)$$
$$g(f) = 44.49 \cdot \log(f) + 4.78 \cdot (\log(f))^2$$

Parameter	Description	Default Value
a_0	Constant offset in dB. This value is simply added to loss grid. By adjusting this value, the mean error can be minimized. It regulates the absolute level of the loss curve.	36.8
a_1	Distance influence coefficient. Physically it represents loss dependant on distance such as atmospheric (dust, hydrometeors, etc...) losses. It regulates slope of the curve.	30.2
a_2	Transmitter height influence coefficient. It is related to errors in DTM, real Earth curvature, etc. It regulates loss curve vertical position like the a_0 , but with respect to antenna height	-12.0
a_3	Okumura-Hata type of multiplying factor for $\log(h_B)\log(d)$	0.1

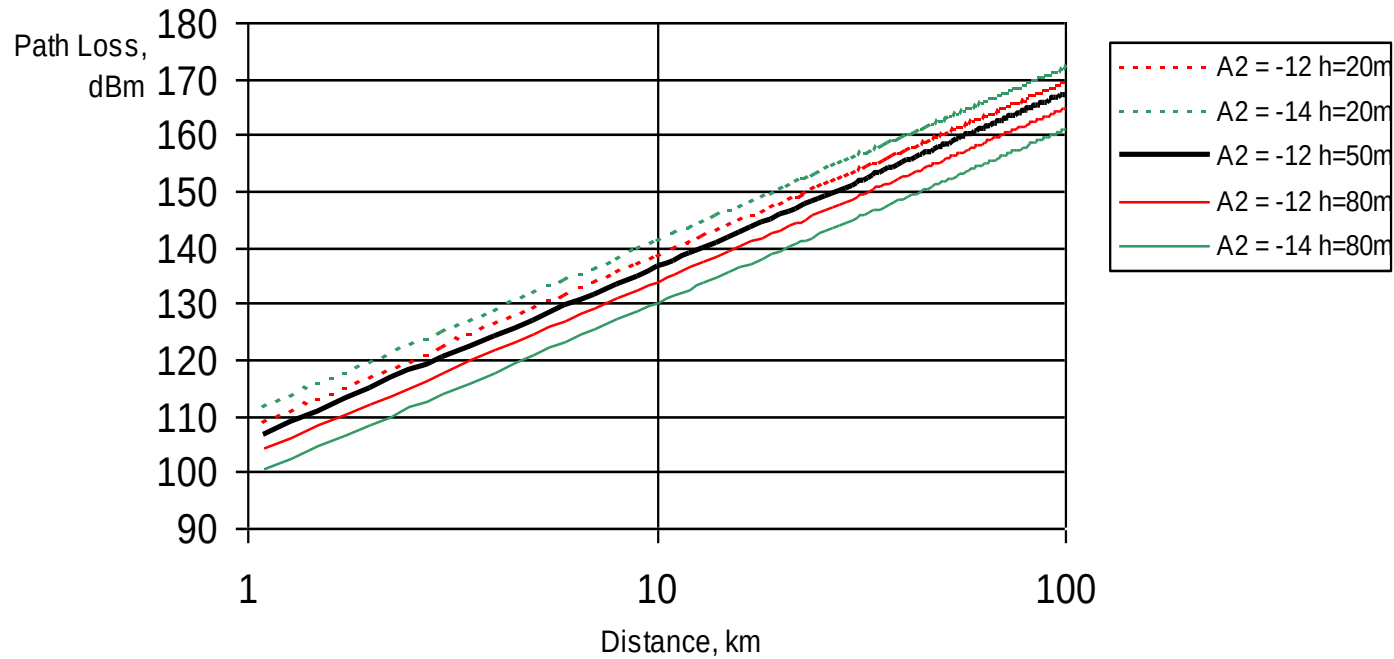
- **9999 Model** is very convenient for calibration
- Empirical parameters **a0-a3** can be deduced from the measured path loss dependence on distance – drive-tests
- **a0** is a constant offset of path loss curve



- 9999 Model is very convenient for calibration
- **a1** regulates slope of the path loss curve

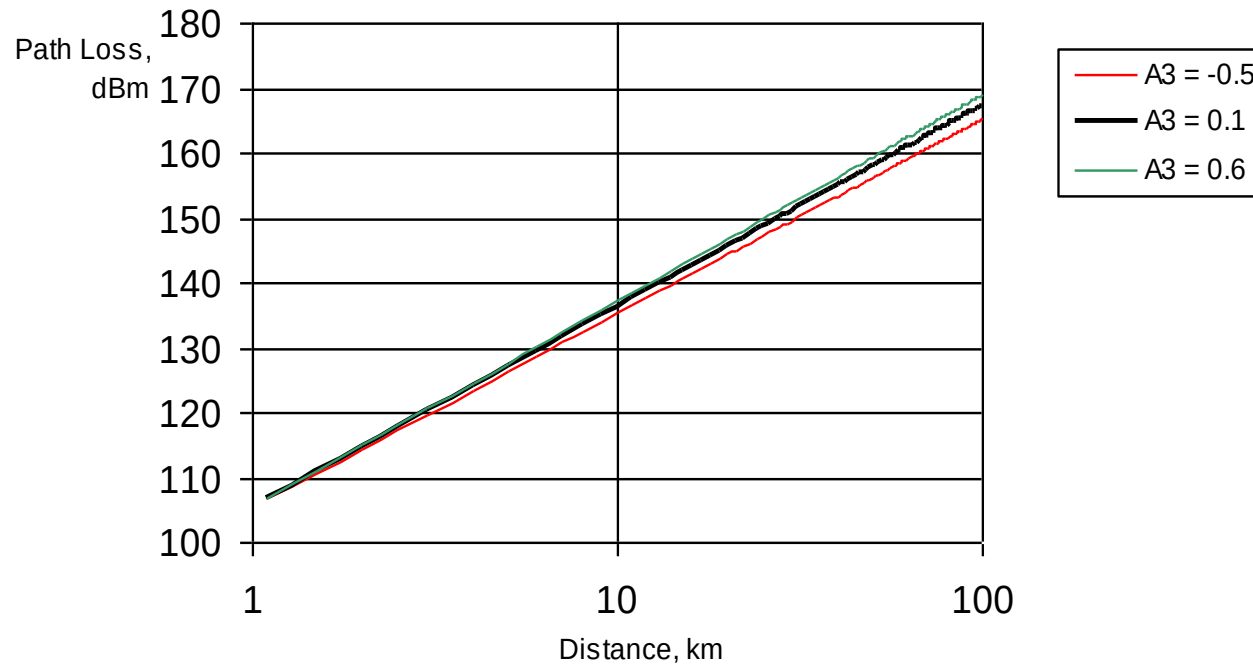


- **9999 Model** is very convenient for calibration
- **a2** regulates loss curve vertical position like a0, but with respect to antenna height



9999 Ericsson: A3

- 9999 Model is very convenient for calibration
- **a3** defines slope of the path loss curve for different base station antenna heights



Path loss equation

Path loss in dB:

$$L = k_{off} + k_{\log D} \cdot \log(d) + k_{\log F} \cdot \log(f)$$

K_{Off} - Constant offset (dBm). Default value **32 (!)**

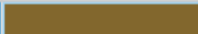
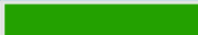
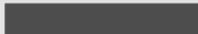
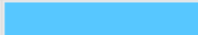
K_{LogD} - Distance influence coefficient. Default value 20

K_{LogF} - Frequency influence coefficient. Default value 20

Offset coefficient, dB:	32
Distance coefficient near:	20
Distance coefficient far:	40
Frequency coefficient:	20

- Diffraction loss for solid obstacle:
 - Building clutter class
 - Elevation
- Clutter loss
 - Based on diffraction calculation
 - P.2108 Clutter Loss
- Penetration loss (Outdoor – Indoor)
- Receiver loss

▼ Clutter class: Buildings

Clutter name	Color
Buildings	
Forest	
Dense forest	
Urban	
Dense urban	
Bare ground	
Crops	
Water	
Road	

Nominal distance, m:

Diffraction loss coefficient:

Penetration loss offset, dB:

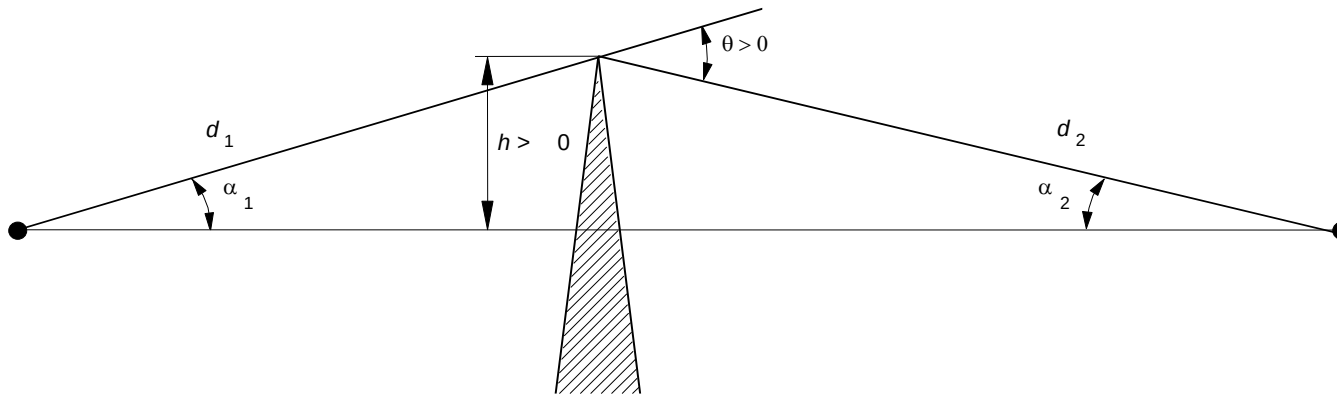
Penetration loss distance coefficient:

Penetration loss frequency coefficient:

Receiver point loss offset, dB:

Rec. ITU-R P.526

Idealized model of diffraction over a single obstruction.



P.2108 Clutter Loss Estimation

- Method 1: Additional clutter shadowing loss with diffraction as dominant effect (section 3.1)

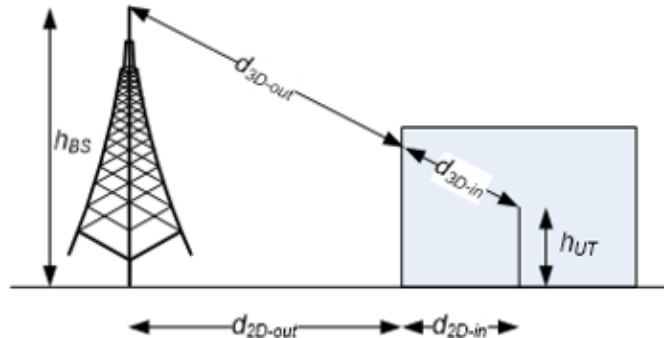
Input parameters:				
Frequency, GHz (0,03-3 GHz):	2.1			
Antenna height (agl), m:	1.5			
Urban scenario street width, m:	27 (default value 27 m)			
Representative clutter height, m:	10			
Clutter clearance angle, deg:	17.47477426			
v :	6.04019561			
$J(v)$:	28.45751028			
Adjusted $J(v)$:	28.45751028			
Clutter loss for open area scenarios:		22.4 dB		
Clutter loss for obstructed area scenarios:		19.6 dB		

clutter

Field:	Selection:	Rows:		
OBJECTID *	clutter_class_name	clutter_class_id	height	nominal_distance
1	water	1	0	27
2	trees	2	20	27
3	flooded vegetation	4	3	27
4	crops	5	2	27
5	built area	7	2	27
6	bare ground	8	0	27
7	snow	9	0	27
8	rangeland	11	3	27
Click to add new row.				

Penetration loss (Outdoor – Indoor)

CE Outdoor to Indoor Path Loss calculation is realised based on method recommended in 3GPP TR 38.901 ([ref URL](#)). This method accounts for indoor portion of the total radio signal propagation path as shown in picture:



Definition of indoor propagation path in 3GPP TR 38.901

For general purpose modelling of typical building entry losses, two types of loss profiles are assumed:

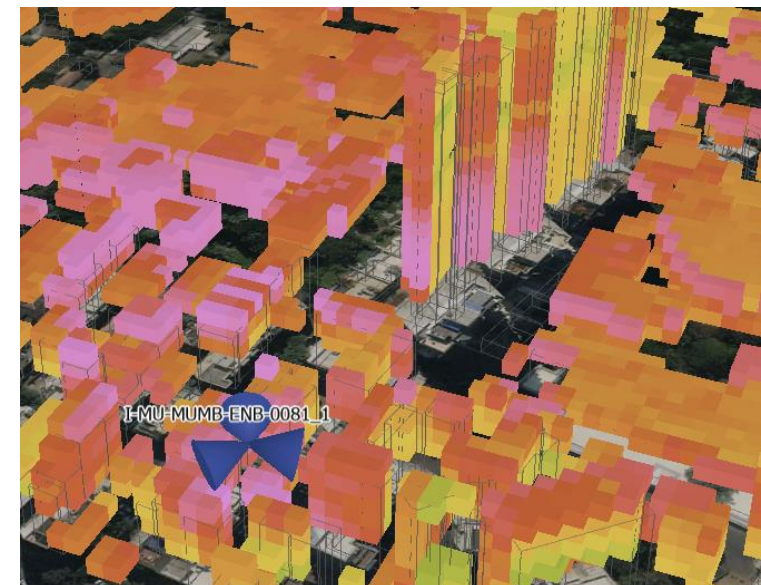
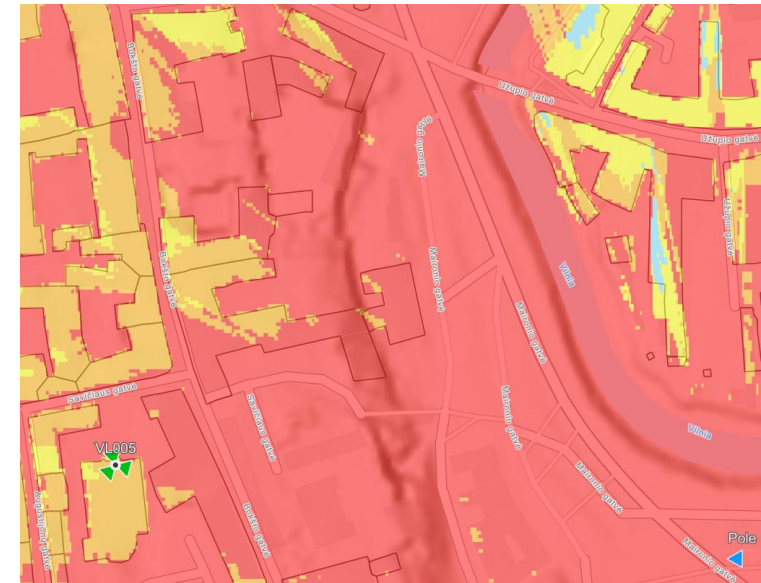
- Low-loss BEL Model assumes a wall penetration losses characteristic of average traditional buildings;
- High-loss BEL Model assumes a wall penetration losses characteristic of modern thermally insulated buildings.

The corresponding BEL and building penetration losses are calculated as follows:

	Path loss through external wall: PL_{tw} in [dB]	Indoor loss: PL_{in} in [dB]	Standard deviation: σ_P in [dB]
Low-loss model	$5 - 10 \log_{10} \left(0.3 \cdot 10^{\frac{-L_{glass}}{10}} + 0.7 \cdot 10^{\frac{-L_{concrete}}{10}} \right)$	$0.5 d_{2D-in}$	4.4
High-loss model	$5 - 10 \log_{10} \left(0.7 \cdot 10^{\frac{-L_{IRglass}}{10}} + 0.3 \cdot 10^{\frac{-L_{concrete}}{10}} \right)$	$0.5 d_{2D-in}$	6.5

Where:

$$L_{glass} = 2 + 0.2f \quad L_{concrete} = 5 + 4f \quad L_{IRglass} = 23 + 0.3f \quad f - \text{frequency in GHz.}$$



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Thank you!

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